

CME 222

MASS TRANSFER



Diffusive Mass Transfer

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Table of Content

- Concentrations, Velocities
- Fluxes & Fick's First Law

Concentrations, Velocities

Concentration

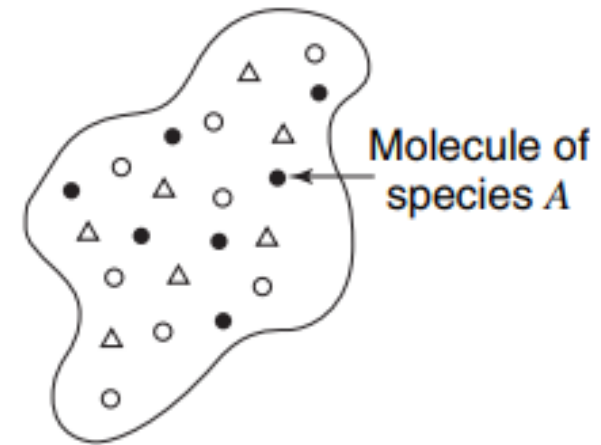
Mass Concentration

$$\rho_A = \frac{m_A}{V}$$

ρ_A = mass concentration of species A

m_A = mass of species A

V = total volume of mixture



Total Mass Concentration/Density

$$\rho = \sum_{i=1}^n \rho_i$$

Concentration

Specific Volume (v)

- Volume occupied by a unit mass of the substance.
- Inverse of density

$$v = \frac{V}{m} = \frac{1}{\rho}$$

Mass Fraction (w)

$$w_A = \frac{m_A}{m} = \frac{\rho_A}{\rho}$$

m_A = mass of A
 m = total mass of mixture

$$\sum_{i=1}^n w_i = 1$$

Summation of mass
fraction equal to 1

Concentration

Molar Concentration

- Mole of A per unit volume of mixture

$$c_A = \frac{n_A}{V} = \frac{\rho_A}{M_A}$$

- For gas phase, concentration often expressed in term of pressure

$$c_A = \frac{n_A}{V} = \frac{P_A}{RT}$$

$$\sum_{i=1}^n c_i = c$$

n_A = mole of A

P_A = partial pressure of A (mole fraction x total pressure)

M_A = molecular weight of A

Concentration

Mole Fraction

For solid and liquid,

$$x_A = \frac{n_A}{n} = \frac{c_A}{c}$$

For gas,

$$y_A = \frac{n_A}{n} = \frac{c_A}{c} = \frac{P_A}{P}$$

Summation of mole fraction equal to 1

Concentration

How can I correlate
mass fraction and
mole fraction?



$$x_A \text{ or } y_A = \frac{\frac{w_A}{M_A}}{\frac{w_A}{M_A} + \frac{w_B}{M_B}}$$

$$w_A = \frac{x_A M_A}{x_A M_A + x_B M_B}$$

Can be applied for y_a

Example 1

The composition of air is often given in terms of only two principal species in the gas mixture

$$\text{O}_2, y_{\text{O}_2} = 0.21$$

$$\text{N}_2, y_{\text{N}_2} = 0.79$$

The molecular weight of oxygen is 0.032 kg/mol and of nitrogen is 0.028 kg/mol. Condition: 25 °C and 101300 Pa abs.

Determine

- i. Mean molecular weight of the air
- ii. Mass fraction of both oxygen and nitrogen
- iii. Molar concentration of both oxygen and nitrogen

Velocity

- In multicomponent mixture, each species will normally move at different velocities

Mass-Average Velocity

- Defined in term of mass densities

$$\mathbf{v} = \frac{\sum_{i=1}^n \rho_i \mathbf{v}_i}{\sum_{i=1}^n \rho_i} = \frac{\sum_{i=1}^n \rho_i \mathbf{v}_i}{\rho} = \sum_{i=1}^n \omega_i \mathbf{v}_i$$

Velocity

Molar-Average Velocity

- Defined in term of molar concentration

$$\mathbf{V} = \frac{\sum_{i=1}^n c_i \mathbf{v}_i}{\sum_{i=1}^n c_i} = \frac{\sum_{i=1}^n c_i \mathbf{v}_i}{c} = \sum_{i=1}^n x_i \mathbf{v}_i$$

Velocity

Diffusion Velocity

Velocity of a species in relative to average velocity

$v_{id} = v_i - v$ Diffusion velocity of species i relative to mass-average velocity

$V_{id} = v_i - V$ Diffusion velocity of species i relative to molar-average velocity

Example 2

A gas mixture containing 65 mol% NH_3 , 8 mol% N_2 , 24 mol% H_2 and 3 mol% Ar is flowing through a pipe 25 mm in diameter at a total pressure of 4 atm.

The velocities of the components are as follows:

$$\text{NH}_3 = 0.03 \text{ m/s}$$

$$\text{N}_2 = 0.03 \text{ m/s}$$

$$\text{H}_2 = 0.035 \text{ m/s}$$

$$\text{Ar} = 0.02 \text{ m/s}$$

Calculate the molar-average, mass-average velocity, and mass diffusion velocity of H_2 .

Fluxes & Fick's First Law

Fick's Law - Flux

- In 1855, Fick quantified the observation of diffusion
- Flux is the **amount** of the particular species passes **through** a cross section **area** at a given time (can be in mass or molar unit)
- Fick's first law proposed **molar flux** for isothermal, isobaric system

$$\mathbf{J}_A = -D_{AB} \nabla c_A$$

- For diffusion of a binary mixture of A and B in z-direction

$$J_{A,z} = -D_{AB} \frac{dc_A}{dz}$$

$J_{A,z}$ = molar flux of component A (mol/m².s)

D_{AB} = diffusivity or diffusion coefficient (m²/s)

$\frac{dc_A}{dz}$ = concentration gradient (C_A in mol/m³, and z in m)

Fick's Law - Flux

Can be expressed in another form

$$J_{A,z} = -cD_{AB} \frac{dy_A}{dz}$$

where $c_A = cy_A$

Fick's Law - Flux

$$j_{A,z} = -\rho D_{AB} \frac{d\omega_A}{dz}$$

$$j_{A,z} = -D_{AB} \frac{d\rho_A}{dz}$$

$j_{A,z}$ = mass flux of A (kg/m².s)

Example 3

A mixture of He and N₂ gas is contained in a pipe at 298 K and 1 atm total pressure. At one end of the pipe the partial pressure of He is 0.6 atm and the partial pressure of He is 0.2 atm at the other end (0.2m apart). Calculate the molar and mass flux of He at steady state if $D_{\text{He-N}_2}$ is $0.687 \times 10^{-4} \text{ m}^2/\text{s}$.

Example 4

A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at 700 °C. If a condition of steady state is achieved, calculate the mass diffusion flux of carbon through the plate if the concentration of carbon at position of 5 and 10 mm beneath the carburizing surface are 100 and 66.67 mol/m³, respectively. Assume a diffusion coefficient of 3×10^{-7} cm²/s at this temperature.

Flux

- Flux is contributed by:
 - ✓ Concentration gradient
 - ✓ Bulk motion
- For binary system with constant average velocity, molar diffusion flux can be expressed as

$$J_{A,z} = c_A (V_{Ad,z}) = c_A (v_{A,z} - V_z)$$

$V_{Ad,z}$ is the molar diffusion velocity

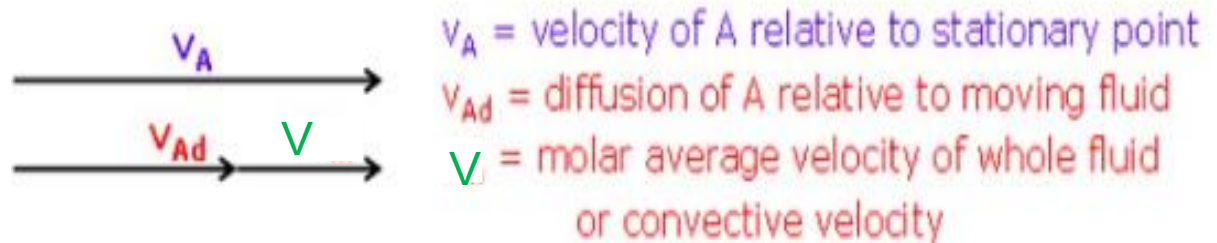
$v_{A,z}$ is the velocity of component A

V_z is the molar-average velocity of stream

Flux

- Diffusion velocity is measured relative to the moving fluid
- To a stationary observer, A is moving faster than the bulk of the phase since its diffusion velocity (V_{Ad}) is added to that of the bulk phase (V)

$$v_A = V_{Ad} + V$$



Multiply with c_A

$$c_A v_A = c_A V_{Ad} + c_A V$$

Flux

Diffusion flux of A relative to moving fluid (**concentration gradient contribution**)

$$c_A v_A = c_A V_{Ad} + c_A V$$

Total flux of A relative to stationary point

Convective flux of A relative to stationary point (**bulk motion contribution**)

Let $N_A = \text{total flux} = c_A v_A$

Flux

Introduce new term total flux represented by N_A

$$J_{A,z} = c_A (v_{A,z} - V_z) = -cD_{AB} \frac{dy_A}{dz}$$

Rearrangement,

$$c_A v_{A,z} = -cD_{AB} \frac{dy_A}{dz} + c_A V_z$$

Molar-average velocity,

$$V_z = \frac{1}{c} (c_A v_{A,z} + c_B v_{B,z})$$

$$c_A V_z = y_A (c_A v_{A,z} + c_B v_{B,z})$$



Flux


$$c_A v_{A,z} = -c D_{AB} \frac{dy_A}{dz} + y_A (c_A v_{A,z} + c_B v_{B,z})$$
$$N_A = c_A v_A$$
$$N_B = c_B v_B$$

$$N_{A,z} = -c D_{AB} \frac{dy_A}{dz} + y_A (N_{A,z} + N_{B,z})$$

This relation may be generalized and written in vector form

$$N_A = -c D_{AB} \nabla y_A + y_A (N_A + N_B)$$

If species A were diffusing in a multicomponent mixture,

$$N_A = -c D_{AM} \nabla y_A + y_A \sum_{i=1}^n N_i$$

where D_{AM} is the diffusivity of A in the mixture

Flux

Expression for mass flux

$$\mathbf{n}_A = -\rho D_{AB} \nabla \mathbf{w}_A + w_A (\mathbf{n}_A + \mathbf{n}_B)$$

where

$$n_A = \rho_A v_A$$

$$n_B = \rho_B v_B$$

Flux

Flux	Gradient	Fick rate equation
$\mathbf{n}_A$	$\nabla \omega_A$	$\mathbf{n}_A = -\rho D_{AB} \nabla \omega_A + \omega_A (\mathbf{n}_A + \mathbf{n}_B)$
	$\nabla \rho_A$	$\mathbf{n}_A = -D_{AB} \nabla \rho_A + \omega_A (\mathbf{n}_A + \mathbf{n}_B)$
$\mathbf{N}_A$	∇y_A	$\mathbf{N}_A = -c D_{AB} \nabla y_A + y_A (\mathbf{N}_A + \mathbf{N}_B)$
	∇c_A	$\mathbf{N}_A = -D_{AB} \nabla c_A + y_A (\mathbf{N}_A + \mathbf{N}_B)$

Consider a dye-filled balloon dropped into the moving stream

- Dye diffuse radially while being carried downstream
- Two contribution in mass transfer

Take Home Message

- Mass concentration vs Molar concentration
- Diffusion velocity (Mass/molar basis)

$$v_{id} = v_i - V$$

- Fick's first law

$$J_{A,z} = -D_{AB} \frac{dc_A}{dz}$$

- Diffusion flux and convective flux

$$c_A v_A = c_A V_{ad} + c_A V$$

$$N_A = -cD_{AB} \nabla y_A + y_A (N_A + N_B)$$